

INEEDI VENKATA SAI ANIRUDH

DEEP LEARNING / MACHINE LEARNING RESEARCH ENGINEER | PYTORCH | COMPUTER VISION | VLMs

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GitHub: Freak205 | LinkedIn: Venkata Sai Anirudh Ineedi | ForgeLens Repository

PROFILE

Final-year Computer Science Engineering undergraduate focused on deep-learning research for document images. Independently owned ForgeLens end-to-end: PyTorch training and debugging, pixel-level localization, leakage-safe data pipelines, calibration, robustness testing, and open-weight SmolVLM2 LoRA fine-tuning. Also ships tested, user-facing software.

TECHNICAL SKILLS

Deep Learning & Vision: PyTorch, torchvision, CNNs, U-Net, image classification, pixel-level segmentation, document-image forensics, residual features, CUDA

Transformers & Fine-Tuning: Hugging Face Transformers, Vision-Language Models, SmolVLM2-2.2B, PEFT/LoRA, supervised fine-tuning, completion-only loss, gradient checkpointing

Research & Evaluation: calibration, selective prediction, abstention, robustness testing, generator shift, ROC/PR-AUC, bootstrap confidence intervals, ECE, Brier score, pixel IoU

Systems & Programming: Python, PowerShell, mixed precision, gradient accumulation/clipping, checkpointing, deterministic seeds, Kaggle GPU, Pydantic, pytest, mypy, Ruff, Git/GitHub

FEATURED DEEP LEARNING RESEARCH PROJECT

ForgeLens - Calibrated Multimodal Document Forensics | github.com/Freak205/FORGELENS

Independent research & engineering project | PyTorch, CUDA, Transformers, SmolVLM2, LoRA/PEFT, Hugging Face, Kaggle

- Designed and delivered an end-to-end PyTorch research platform for document-image forgery-risk classification and pixel-level tamper localization, spanning problem framing, data provenance, model training, evaluation, calibrated scoring, abstention, and user-facing inference.
- Built revision-pinned, checksum-backed pipelines for CORD v2, a 2,000-sample balanced copy-move benchmark, and a 1,966-sample paired AIForge/CORD experiment; preserved authentic/forged source groups across leakage-safe train, validation, and test splits.
- Implemented and trained three joint detector families - compact RGB CNN, skip-connected U-Net, and fixed Laplacian/Sobel residual U-Net - using mixed precision, gradient accumulation, deterministic seeds, atomic checkpoints, non-finite-loss detection, and recorded hashes.
- Built an evaluation harness spanning ROC/PR-AUC, bootstrap confidence intervals, calibration, risk-coverage, pixel IoU, nine noisy/capture-corruption tests, latency, and peak VRAM; established checkpoint-rejection gates and rejected four RTX 2050 models when results were statistically or operationally unreliable.
- Fine-tuned the open-weight SmolVLM2-2.2B-Instruct VLM using PEFT/LoRA supervised fine-tuning on a Kaggle Tesla T4; completed one epoch and 16 optimizer steps at ~7 GiB peak VRAM and improved balanced accuracy from 0.500 to 0.547 on a deterministic 64-example public CORD proxy test.
- Delivered 2.45 ms median batch-one local inference, a Pydantic output contract and localhost mask UI; maintained 51 passing tests, strict mypy, Ruff, pinned dependencies, PowerShell automation, DPAPI-secured credentials, a technical report, model card, failure gallery, and reproducibility guide.

INTERNSHIP EXPERIENCE

Full-Stack Developer Intern - Southeast Media | June 2026 - Present

Live: <https://southeastmedia.in> | Next.js 16, React 19, TypeScript, Tailwind CSS, GSAP, Firebase App Hosting

- Owned end-to-end delivery of a live CGI/VFX studio platform spanning seven art-directed business verticals, translating brand direction into a production-ready, responsive web experience.
- Engineered a modular motion and media system using GSAP ScrollTrigger, Lenis, Framer Motion, CSS 3D, canvas noise, and matter.js while preserving reduced-motion accessibility.
- Hardened deployment with Zod validation, rate limiting, honeypot protection, security headers, environment-aware SEO, AVIF/WebP delivery, lazy video playback, caching, and dynamic imports.

Full-Stack Developer Intern - Boltzmann Labs | May 2026 - Present

CLINVARA | Live: <https://www.clinvara.global> | Next.js, TypeScript, Firebase, Firestore, Firebase Functions

- Building CLINVARA, a clinical-skincare ecommerce platform, across frontend architecture, authentication, Firestore data design, serverless functions, automation, SEO, deployment readiness, and documentation.
- Developed an internal operations console for workflow management, lifecycle visibility, content governance, service handling, platform controls, and future channel expansion.
- Implemented resilient environment and fallback handling, canonical routing, sitemap/robots, Open Graph, JSON-LD structured data, and automated social-content ingestion.

EDUCATION

B.Tech - Computer Science Engineering | GITAM University, Hyderabad | Final Year | 2023 - 2027